Kandice Lu

413-834-8510 kandicelu3.14@gmail.com **Portfolio:** https://kandicelu.github.io/

Education

Cornell University – B.S. in Mechanical Engineering **May 2026**

- GPA: 4.11/4.3, 3.93/4.0 — Awards: *Sibley Prize*

Cornell University – M.Eng in Aerospace Engineering **Expected Dec 2026**

Relevant Coursework

- Finite Element Analysis, Vibrations, Material Mechanics, Composites, Aeronautics, Flight Dynamics, Fluid Mechanics, Thermodynamics, Heat Transfer, Mechatronics, System Dynamics, Engineering Computation

Skills

- **Technical Skills:** CAD, FEA, Structural Design and Analysis, Data Analysis, Machining, Welding, DFMA, GD&T
- **Software:** ANSYS, Solidworks, Fusion 360, STAAD.Pro, LabVIEW
- **Programming Languages:** MATLAB, C/C++, Python

Engineering Experience

Full Team Lead, Fabrication Lead **Sep 2023 – Present**
Cornell Steel Bridge Project Team *Ithaca, NY*

- Led a multi-disciplinary team of 30 students through the full design-build cycle of a 23-foot steel bridge, competing on a combined score of weight, construction time, and deflection under a 2,500 lb load, placing third in a regional competition
- Designed the bridge's base geometry and ran FEA across multiple configurations and section properties to optimize the weight-stiffness tradeoff, improving projected structural efficiency by 15%
- Designed a new connection geometry to improve rigidity and assembly speed, validating with CAD and FEA to ensure structural integrity under worst-case loads
- Reviewed all components for fit with adjoining parts, manufacturability, ease of assembly, and compliance with rules
- Directed the team's machining operations: sourced materials, developed procedures for efficient bulk fabrication, and trained junior teammates
- Led static load testing of the final assembly and designed and implemented fixes that reduced deflection by 50%

Undergraduate Researcher **May 2025 – Dec 2025**
ZT Group *Ithaca, NY*

- Developed and optimized fabrication processes for polymer-based thermal interface materials
- Characterized thermal properties with laser flash testing to guide iterative sample creation

Coursework Projects

Finite Element Analysis: Wing Design **Oct 2025 – Dec 2025**

- Designed an airplane wing to minimize weight while meeting deflection and safety factor requirements
- Analyzed and optimized various design configurations in ANSYS, reducing initial weight by over 20%
- Performed manual and automatic mesh refinement tests to ensure convergence and verification of results

Fluids Lab: Wind Turbine Blade Design **Oct 2025 – Dec 2025**

- Designed and tested a small scale wind turbine
- Constructed a mathematical model of blade performance in MATLAB
- Validated model and physical turbine performance through wind tunnel testing

Introduction to Aeronautics: Glider Project **Sep 2024 – Dec 2024**

- Designed a slow flying balsa glider, performing lift and drag calculations to achieve balanced flight
- Modeled glider skeleton with CAD and constructed glider via laser cutting and hand assembly
- Tested and rebalanced constructed glider to achieve desired speed and stability, achieving 5th place in final competition

Work Experience

Finger Lakes Reuse **June 2025 – Aug 2025**
Administrative Project Intern *Ithaca, NY*

- Researched the viability of and assisted in the implementation of transitions to new report creation and price management systems
- Assisted chief officers in creating and organizing various strategic documents, databases, and reports